Incoming request

Read primary cache
refresh=1 skips the read

hit

Cache hit
return response

miss or refresh

Lock applicable?
enabled, cacheable, Redis

no

No lock path
invoke1 as before

yes

Try acquireLock
SET NX EX, unique value

acquired

held

Normal: re-read cache
hit: unlock and return

miss or refresh

invoke1 backend call
writes cache on success

finally: releaseLock
Lua, then response logic

waitForCache loop

Cache hit
return it

Timeout
throws error

Lock owner path

Waiter and cache-hit path

Loop until deadline
TTL + 1 second

Is the lock present?
isLockPresent (EXISTS)

free

held

Normal: read cache
hit returns; refresh skips

miss

Try acquireLock
only the winner proceeds

won

Owner: executeUnderLock

lost

Read cache anyway
serves readers behind a refresh

hit

Return cached

miss

Sleep 100 ms, loop again

after deadline

Deadline reached
throws timeout error

Lock owner path

Cache-hit path